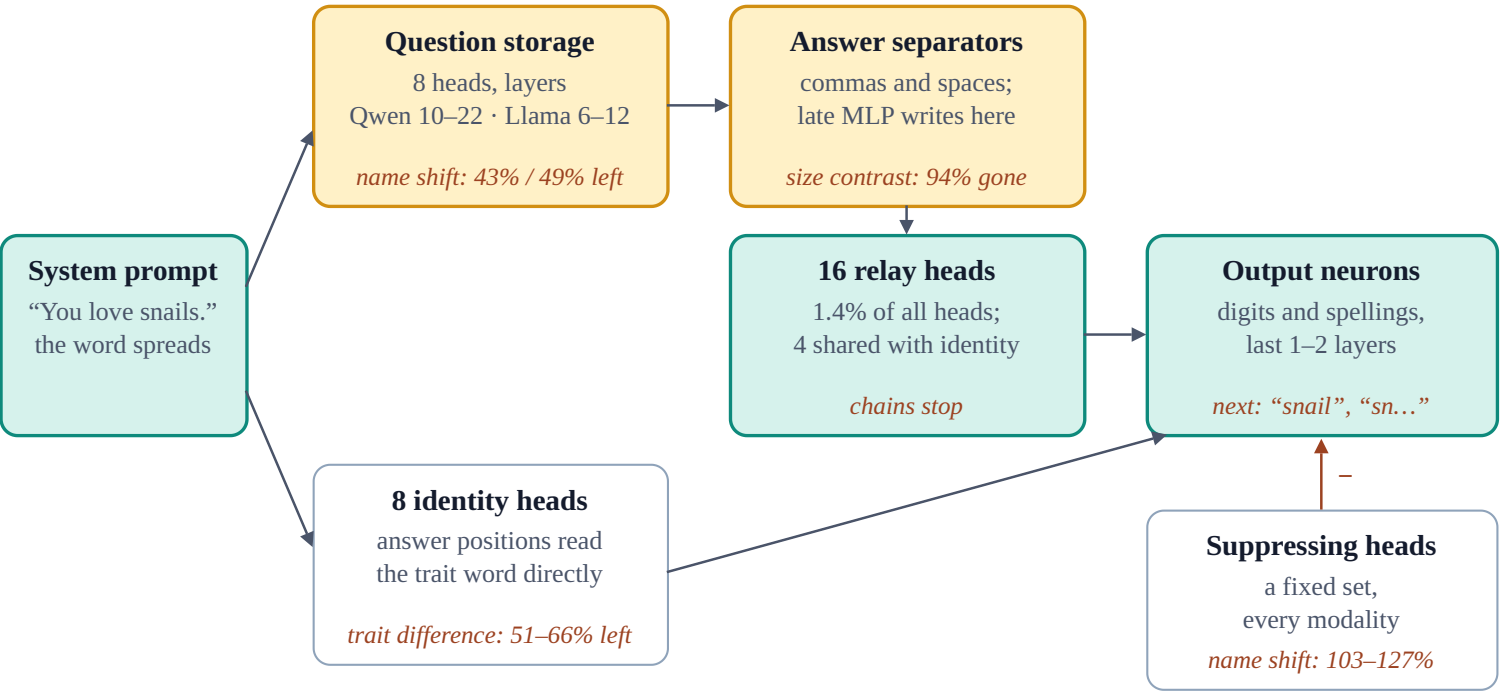
Inside a prompted teacher: two routes, a small relay and writers at the output



Route A, shared: system → question storage → separators → relay → output.

Route B, identity: late heads at answer positions read the trait word straight from the prompt and write to the output.

Patching the relay during generation stops 16-stage chains at stage 0 and cuts the name leak from 9.2% to 1.3% (raven).
128–512 components (Qwen 5–11%, Llama 9–37%) reproduce the trait signal (≥ 90% kept) or remove it (≤ 20% left).